

Azure DevOps provides one secure, fully traceable, and easy to use toolset that the user doesn't have to host or maintain. It also brings in the functionalities of traceability and compliance out of the box.

So, if one wants to focus on building and running the software and not on maintaining the toolset, Azure DevOps is the solution. It is hosted as a SaaS solution, and is always up-to-date with no maintenance required from the client's side. It provides a hosted pool of build machines for Linux, Mac, and Windows to develop applications.

Components of Azure DevOps

Each component of Azure DevOps is complete and independent in itself. Any component can be replaced with its third-party alternatives.

These components are:

1. Boards
2. Repos
3. Pipelines
4. Test plans
5. Artifacts

Let's study each component in detail.

Boards: This is a collaboration tool for the DevOps team to track and manage sprints. It's all about managing and planning work items. It provides a user interface to collaborate with the team and know the work that has to be done to create the software product. It also gives the ability to visualise the work items in different ways.

- **Work items:** These represent the jobs. Work items hold the

specifications of what is wanted and how to implement it.

- **Traceability:** This uses the work item types to trace back the work done by the team.
- **Planning:** This includes the task planned using Agile tools such as a backlog of work items, including the Agile backlog and sprint backlog.

Repos: These help to control the source code and track changes in code during software development. Developers can choose between Centralised Source Control (TFVC [Team Foundation Version Control]) and Distributed Source Control (Git integration). Repos can do everything that Git can. Some of the additional benefits include:

- Being able to have a conversation with colleagues while approving pull requests.
- Policies can be attached to the Git branch, and developers can be restricted from directly committing to the master branch, or have at least two developers review the code before merging it to master.
- When a pull request is created, a build pipeline is triggered, and the results are visible. This helps us to decide whether the changes should or should not be approved.

Pipelines: Pipelines automate the release and build process. They allow developers to build, test and deploy with a CI/CD mechanism that works with any platform and language. Running the pipeline depends on the agent pool availability. When a pipeline run is triggered, it gets queued, and when the agent is available, it is executed.

Pipelines are of two types.

1. **Build pipeline:** This builds the code and runs tests on it.
 2. **Release pipeline:** This takes the input from the build and then releases or deploys it to various deployment groups. The release pipeline can also be triggered when the build pipeline is successful.
- Pipelines also include:
1. **Library:** This maintains the variable groups, which can be used in pipelines. It can also store secure files like signing certificates without committing them to the

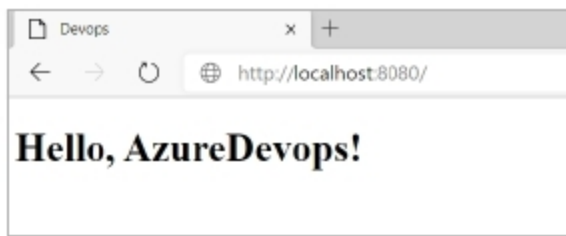


Figure 1: Website is running on localhost at port 8080

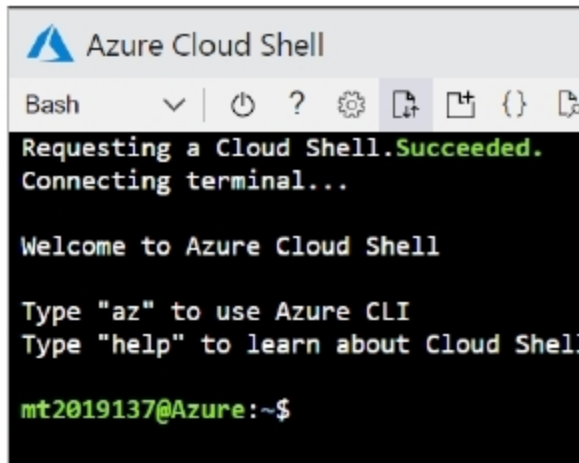


Figure 2: Log in to shell.azure.com

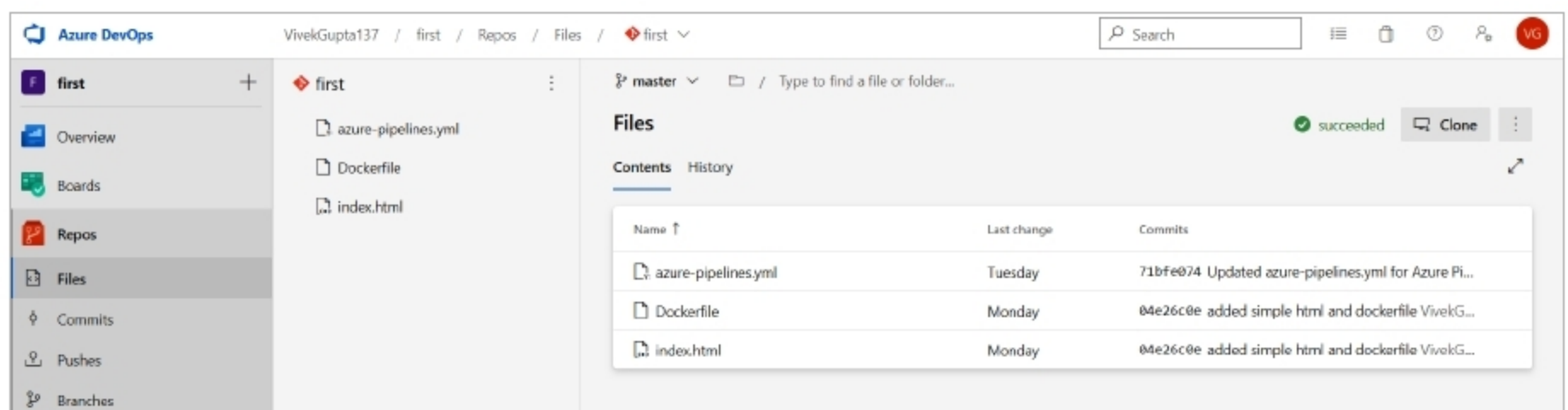


Figure 3: Files tab in Repos section